Multi-level Analysis Of The Stability-Flexibility Dilemma

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Abstract

A dominant view in cognitive control research concerning stability and flexibility is that of a tradeoff relationship, where the more cognitively stable a task representation becomes the harder it is to flexibly switch to another task. Although widely replicated, this tradeoff account is typically supported using a single level of analysis of the experimental effects of switch cost as a degree of conflict blockwise (Goschke 2000b, 2000a; Rogers and Monsell 1995; Egner 2007). A smaller body of research supports a different account of stability-flexibility, where on the level of within-subject dynamics stability and flexibility are found to co-occur (Mayr, Kuhns, and Hubbard 2014). This raises the question that the stability–flexibility tradeoff seen through experimental level effects is not a fundamental property of cognitive control but may instead reflect the analytic level at which it is measured. Moreover, individual differences work hints at substantial variability in how people balance stability and flexibility, suggesting that the relationship may not be uniform across all individuals (Mayr and Grätz 2024).

The present study investigates where differences in the stability-flexibility dilemma arise through a multi-level analytic approach. Additionally, we take a more direct and sensitive approach to understanding cognitive control mechanisms through looking at the degree of control required pre- and post- task-switching. Across two experiments, we examined stability–flexibility dynamics at three analytic levels: experimental effects, within-subject dynamics, and individual differences. We find that it indeed matters to what level stability/flexibility are analyzed. Evidence for the co-occurrence of tradeoff and antitradeoff relationships are found on different levels of analysis, under-

scoring the importance of a multi-level approach when interpreting stability-flexibility interactions. By showing that stability–flexibility dynamics vary across analytic scales, our results challenge the presumed ubiquity of the dominant tradeoff view and highlight the necessity of multi-level approaches for understanding the mechanisms that govern cognitive control.

Introduction

A dominant view in research on cognitive control holds that adjustments of control are constrained by a general stability-flexibility tradeoff. Specifically, increasing the stability of a task-relevant representation makes flexible changes of that representation more difficult (Goschke 2000b). You may be planning to give a speech for example, relying on your notes helps you stay on message reflecting high stability, but it can also make it harder to smoothly switch into an improvised, conversational mode when a question arises thereby reducing flexibility.The very thing that keeps you anchored can also slow you down when you need to pivot. Understanding stability and flexibility mechanisms is also of vital importance in the clinical world, with impaired regulation of cognitive control being implicated in various neurodevelopmental disorders such as attention deficit hyperactivity disorder, cerebral palsy, and autism spectrum disorder (Cepeda, Cepeda, and Kramer 2000; D'Cruz et al. 2013; Tajik-Parvinchi et al. 2021). In these cases, disruptions in cognitive control can manifest as either overly rigid behavior or excessive distractibility, underscoring the importance of understanding the mechanisms that support both maintaining and updating task-relevant representations.

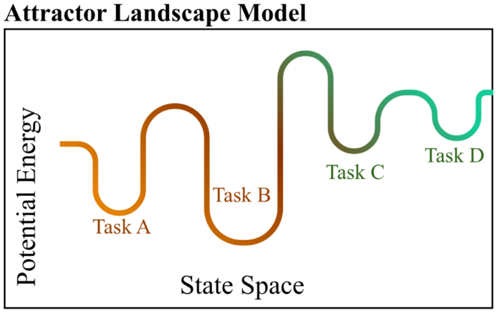
Despite this importance, most empirical work has examined stability–flexibility dynamics using blockwise manipulations, contrasting "stable" versus "flexible" blocks, and has evaluated the tradeoff primarily at the experimental-effects level, averaging across trials and participants (Rogers and Monsell 1995; Goschke 2000a; Egner 2007). This approach has been instrumental in establishing the classic tradeoff pattern, but it also limits what can be observed. Blockwise designs cannot capture moment-to-moment adjustments that occur as people move from one trial to the next, nor can they reveal whether stability and flexibility might co-occur within individuals under certain conditions. As a result, the field may be missing critical dynamics that unfold at finer temporal scales.

To address this limitation, we take a more sensitive approach to capture switch-cost. Previous to current trial control demands offer a more detailed and sensitive account of shifts in cognitive control,determining switch-cost instead as a function of the degree of control demands on previous trial switch and post trial switch manipulated independently. Manipulating previous-trial switch and post-switch demands independently provides a more detailed and responsive measure of how stability and flexibility interact on a trial-by-trial basis. We then examine these dynamics across three complementary levels of analysis: (1) experimental level, (2) within-subject dynamics, and (3) individual differences. This multi-level framework allows us to test whether the stability–flexibility tradeoff emerges consistently across analytic scales or whether different levels reveal different patterns. We begin first by introducing the analytic level with the widest body of prior research: experimental effects.

Stability-Flexibility Tradeoff and Experimental Effects.

The prediction of a tradeoff relationship arises from connectionist models of cognitive control. The behavior of these models can be described within an "attractor landscape" (Brown, Reynolds, and Braver 2007; Geddert and Egner 2022). In this framework, different task representations form basins in a state space, and the depth of a basin reflects how strongly a task is currently activated. A deeper basin corresponds to a more stable task state, but it also requires more "energy" to move out of, making task switching more difficult.

Attractor Landscape Model



- I want to make this more understandable for a wider audience. attractor landscape will also have an accompanying figure so that it will be more understandable. I want to explain terms like conflict and task basins so it will be more clear when we move forward with explaining three different levels of analysis.

One data pattern that is consistent with this tradeoff prediction arises when examining task-switch costs as a function of previous-trial conflict. High conflict on the previous trial is assumed to strengthen the currently active task representation, effectively deepening its basin. This increased stability should then make switching on the next trial harder, producing larger switch costs. Indeed, much of the dominant narrative in cognitive control has been built around this tradeoff pattern (Goschke 2000b). Yet, there are also several recent results that call the generalizability of this finding into question (Geddert and Egner 2022; Brown, Reynolds, and Braver 2007; Mayr, Kuhns, and Hubbard 2014).

Stability-Flexibility Anti-Tradeoff Arises When Examining Within-Subject Dynamics.

note: this needs to be built on when interpreting the data more, read the mayr paper that finds co-occurance when analyzing within-sub dynamics in the meantime.

A recent reevaluation of the generalizability of the stability-flexibility tradeoff has posited that this negative tradeoff relationship may occur only in highly specified contexts (Mayr and Grätz 2024).These authors pointed toward a possible anti-tradeoff pattern, implying that cognitive stability and flexibility are both driven by a common factor and therefore can co-occur (Mayr, Kuhns, and Hubbard 2014). This paper adopts a more nuanced approach by examining when stability–flexibility dynamics emerge, weaken, or disappear. Across two experiments we conduct analysis on the experimental effects, within-subject dynamics, and individual differences level to identify the conditions under which tradeoffs arise and when they do not.

Individual Differences

note: we find an antitradeoff occuring, need to interpret the results for this section more too. talk about age as a factor or not necessary for this paper? maybe that can come in the discussion portion about other individual dif factors that we can explore with this level of analysis.

extra stuff

Cognitive Map Prediction Model. Contrary to predictions from the attractor model, there are recent hints suggesting the possibility of an anti-tradeoff pattern, where flexibility and stability appear to co-occur (Mayr, Kuhns, and Hubbard 2014; Morales, Moss, and Mayr 2022). In response to these results, Mayr and Graetz have suggested replacing the attractor-landscape perspective with a cognitive map conceptualization (Fig. 1B). Here, different tasks are locations within a task space that can be represented with varying degrees of resolution. The higher the resolution, the more stable tasks are encoded, but also the easier they are to find when switching, resulting in an anti-tradeoff pattern (Mayr and Grätz 2024). Despite some evidence for such anti-tradeoff patterns (Morales, Moss, and Mayr 2022), this prediction has not been rigorously tested.

Experimental Effects. Here we test whether experimental manipulations of previous and current trial control demands interact with the switch contrast in a manner that is consistent with a flexibility/stability tradeoff model.

Within-subject Dynamics. Here we test whether trial-by-trial RT auto-correlations in combination with considering the experimental effects express a flexibility/stability tradeoff.

Individual Differences. Here we test whether individual differences in measures of "stability/efficiency" and in measures of flexibility are consistent with the tradeoff model. - tobias egner: individucal dif but looks at data in a block-wise manner. found no relationship - study done in lab reference

note: emphasize that we are lookng at stab

switch cost as a degree of conflict blockwise has been the predominiant way of understnadning cog control we look at swtihc cost as a function of the degree of control demands on n-1 switch and post switch manipulated independently. this is a more stringent way of analyzing cog control than predominant tests (tobias egner)

this is a very sentitive test of stab/flex bc we are looking at the degree of control required pre and post switch. this is a more immediate test than previous tests of stab/flex. it is a more sensitive and direct test of stab/flex tradeoff.

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